

GENERAL SPECIFICATION G7.16

Integrated Security Management System (ISMS)

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G7.16 INTEGRATED SECURITY MANAGEMENT SYSTEM (ISMS)

G7.16.1 Scope

(a) General

- (i) The Contractor is to include the facilities as detailed in this specification as follows:
 - 1) Closed Circuit TV System (CCTV);
 - 2) Access Control System; and
 - 3) Fence Intruder Detection & Alarm.
- (ii) Contractor to include for all work as detailed in this Specification to comprise the whole of the labour and all materials to form a complete integrated security management system installation.
- (iii) Contractor to include provision to carry out all tests, adjustments and commissioning as is required to give an effective operational installation.
- (iv) Contractor to include for all Witness Testing with the Client, or his Representative, and training of the Client's Staff in the pricing.
- (v) Contractor is to take responsibility for the design and build of a fully working security management system that meets the requirements of this specification.

(b) Standards and Codes

- (i) Unless otherwise agreed by the S.O., the works are to comply with the relevant requirements of the Standards and Codes listed below or referenced in the body of the specification. Materials or products manufactured in accordance with alternative Standards and Codes may be proposed for approval by the S.O., provided it can be demonstrated that the materials or products comply with the requirements of the specified standards. All Standards and Codes quoted are current version, unless specific year references are noted.
- (ii) Provide and install equipment and components in accordance with the following standards
 - 1) Install all electrical systems in accordance with SS CP5.
 - EN 50131;
 - EN 50132;
 - EN 50133;
 - EN 50136;
 - BS 8591; and
 - NSI NACP 1, 2, 3, 11, 20 and 30 (together with other documents referenced by them), unless more onerous requirements are specified herein, in which case the requirements of this specification shall take precedence.

- 2) Install fibre optic cables in accordance with the requirements and recommendations of BS EN 50174-1, BS EN 50174-2 and BS EN ISO 9000 for Installation of Fibre Optic Cabling.
 - 3) State compliance or non-compliance with:
 - NSI listed in relevant categories, or equivalent local standard; and
 - ISO 9000 approved.
 - 4) Provide supporting documentation in the form of Quality Assurance Certificates and proof of Membership of relevant organisations.
- (c) Local Authorities
- (i) Ensure that the works comply with the relevant authority regulations and directions.
 - (ii) Where the terms “testing authority”, “approving authority”, “relevant authority”, “local authority” or “local authorities” are used in this section, the bodies referred to are as follows:
 - 1) Energy Market Authority (EMA);
 - 2) SP Services Ltd;
 - 3) Fire Safety & Shelter Bureau;
 - 4) National Environment Agency; and
 - 5) SPRING Singapore.

G7.16.2 Closed Circuit TV System (CCTV)

- (a) General
- (i) CCTV system shall be used for picture taking, recording, reproduction, amplification, transmission and distribution of video signals to dedicated receivers or monitors. A CCTV system shall comprise of items for monitoring, transmission and controlling purposes.
 - (ii) Operation of cameras used outdoors must not be affected by lightning surge current or direct heat of lightning. The Contractor shall provide lightning protection system for each outdoor camera and mounting pole.
 - (iii) The Contractor shall, as a minimum, furnish and install all CCTV cameras, mounts, housings, power supply systems, UTP cable, fibre optic cable, network cables, network switches, balun connectors, equipment racks, monitors and all other necessary hardware and software for functioning of the complete system.
 - (iv) CCTV system shall be Internet Protocol (IP) network based digital video system using IP cameras
 - (v) Standard UTP cable, Ethernet-based network cables, fibres and switches shall be utilized for transmission medium and in the distribution network.
 - (vi) IP-based video encoders; if specified to be provided in Particular Specifications, shall be utilized to convert any analogue signal to digital video signals.

- (vii) High performance servers; if specified to be provided in Particular Specifications, shall be provided for system management and recording of the digital video.
 - (viii) Workstation(s) and LCD TVs; if specified to be provided in Particular Specifications, shall be for control and monitoring purposes.
 - (ix) CCTV system shall not be limited to equipment sited at one locality or confined to a single centralized control regime but may include the use of remote transmission or decentralized control. Detailed tailoring of CCTV system shall refer to Particular Specifications.
 - (x) The network structure shall be scalable and modular to cater for future expansion or reduction of the system without having to shift the point of connection or interrupt the system operation. The CCTV system network shall be linked up via cabling as specified and shall operate as a complete integrated system.
 - (xi) The system network and the equipment shall be resilient to provide high availability and to facilitate system maintenance activities.
 - (xii) CCTV system shall be provided with software and hardware security protection against unauthorized modification to the system configuration, setting, viewing and reviewing of any recorded data.
 - (xiii) Viewing of the real-time video and reviewing of the recorded video or both shall not interrupt and affect the capability and performance of the system.
 - (xiv) The hardware equipment offered shall be state-of the-art, standardized commercial off-the-shelf and modular equipment.
 - (xv) The CCTV network communication shall employ an open system protocol.
- (b) System Architecture and Components
- (i) A CCTV system shall typically consist of:
 - 1) Picture generation equipment, generally in the form of colour cameras;
 - 2) A video signal transmission and distribution network connected by fibre optic cable;
 - 3) A central observation suite and other auxiliary observation stations, where the television signals are displayed on monitors;
 - 4) Other ancillary equipment e.g. mixing, splitting or multiplexing equipment to obtain special display effects on monitors;
 - 5) Recording and playback;
 - 6) All Pan-Tilt-Zoom (PTZ) cameras shall be provided with an option to allow the user to -pre-determine the locations that he/ she wants to monitor (i.e. preset locations), including the setting of a routine pattern, where necessary. The CCTV camera system shall have a 'default settings' function, which allows the PTZ cameras to auto reset to their original position after pre-determined time duration; and

- 7) CCTV System Integrator or vendor shall propose standard codec to achieve optimal compression ratios while ensuring no or little loss of image/ video quality.
- (c) Pan-Tilt-Zoom (PTZ) Network Camera Minimum Requirements:
- (i) The network camera shall support open API, standard protocol such as ONVIF (Open Network Video Interface Forum), PSIA (Physical Security Interoperability Alliance), third party Software Development Kit (SDK) and ensuring interoperability between network video products regardless of the manufacturer.
 - (ii) Image Sensor: 1/2.8" Progressive Scan Megapixel CMOS sensor.
 - (iii) Resolution: HD 1080p with 1920x1080 effective Pixels or its equivalent, 2 Megapixels.
 - (iv) Focal Length: 4.5mm~135mm.
 - (v) Angle of View: 67°~2.4°.
 - (vi) Optical Zoom: 30x.
 - (vii) Focus Control: Auto/Manual.
 - (viii) Video Compression: H.265/H.264/MJPEG.
 - (ix) Video Streaming capability: 3 Streams.
 - (x) Video Streaming Resolution: 1080P/720P.
 - (xi) Streaming Methods: Unicast/ Multicast.
 - (xii) Day/Night: True Day/Night (ICR), colour/ black and white.
 - (xiii) Noise Reduction: Digital Noise Reduction (DNR) (2D/3D).
 - (xiv) Audio Compression: Pulse Code Modulation (PCM) (G.711a, G.711u).
 - (xv) Interoperability: PSIA, ONVIF, Computer Generated Imagery (CGI).
 - (xvi) Edge Storage: Network Attached Storage, Local PC/ workstation for instant recoding, Micro SD Memory Slot supporting up to 128GB.
 - (xvii) Web Viewer: IE10/Google/Firefox/Safari.
 - (xviii) Smart Phone: IOS and Android.
 - (xix) Ethernet: RJ45 (10/100Based_T).
 - (xx) Protocol: IPv4/IPv6, HTTP, HTTPS, SSL, TCP/IP, UDP, UPnP, ICMP, IGMP, SNMP, RTSP, RTP, SMTP, NTP, DHCP, DNS, PPPOE, DDNS, FTP, IP Filter, QoS, Bonjour, 802.1x.
 - (xxi) Power Supply: AC23V/1.5A or Power Over Ethernet (PoE) + (802.3at).
 - (xxii) Power Consumption: 13W, 23W or PoE.
 - (xxiii) Horizontal Adjust Angle: 0°~360°; Vertical Adjust Angle: 0°~90°; Auto Flip Angle: 180°.
 - (xxiv) Manual Speed: 0.1°-350°/ second for Pan, 0.1°-250°/ second for Tilt.
 - (xxv) Preset speed: 500°/ second for Pan, 500°/ second for Tilt.

- (xxvi) Operating Temperature: -40°C to +60°C.
- (xxvii) Storage Temperature: -20°C to +70°C.
- (xxviii) Ingress Protection: IP67.
- (xxix) Vandal Resistance: IK10.
- (xxx) Humidity: 0% to 90% RH, non-condensing.
- (xxxi) Anticorrosion standard: WF2, NEMA 4X, C5-M.
- (d) Explosion Proof PTZ Network Camera Minimum Requirements:
 - (i) The Network Camera shall support open API, standard protocol such as ONVIF (Open Network Video Interface Forum), PSIA (Physical Security Interoperability Alliance), third party Software Development Kit (SDK) and ensuring interoperability between network video products regardless of the manufacturer.
 - (ii) Explosion Proof PTZ network camera shall be subjected and certified to rigorous dust and immersion test (IP67). Explosion Proof PTZ network camera shall have IP67 ingress protection rating and are made of 316L stainless steel, corrosion-resistant and suitable for use in hazardous area.
 - (iii) Image Sensor: 1/2.8" Progressive Scan Megapixel CMOS sensor.
 - (iv) Resolution: HD 1080p with 1920x1080 effective Pixels: 1920x1080 or its equivalent, 2 Megapixels.
 - (v) Focal Length: 4.5mm~135mm.
 - (vi) Angel of View: 67°~2.4°.
 - (vii) Optical Zoom: 30x.
 - (viii) Focus Control: Auto/Manual.
 - (ix) Video Compression: H.265/H.264/MJPEG.
 - (x) Video Streaming capability: 3 Streams.
 - (xi) Video Streaming Resolution: 1080P/720P.
 - (xii) Streaming Methods: Unicast/ Multicast.
 - (xiii) Day/Night: True Day/Night (ICR), colour/ black and white.
 - (xiv) Noise Reduction: Digital Noise Reduction (DNR) (2D/3D).
 - (xv) Audio Compression: Pulse Code Modulation (PCM) (G.711a, G.711u).
 - (xvi) Interoperability: PSIA, ONVIF, Computer Generated Imagery (CGI).
 - (xvii) Edge Storage: Network Attached Storage, Local PC/ workstation for instant recoding, Micro SD Memory Slot supporting up to 128GB.
 - (xviii) Web Viewer: IE10/Google/Firefox/Safari.
 - (xix) Smart Phone: IOS, Android.
 - (xx) Ethernet: RJ45 (10/100Based_T).

- (xxi) Protocol: IPv4/IPv6, HTTP, HTTPS, SSL, TCP/IP, UDP, UPnP, ICMP, IGMP, SNMP, RTSP, RTP, SMTP, NTP, DHCP, DNS, PPPOE, DDNS, FTP, IP Filter, QoS, Bonjour, 802.1x.
- (xxii) Power Supply: AC220V (AC100V ~240V) or Power Over Ethernet (PoE) + (802.3at).
- (xxiii) Power Consumption: 40W, 75W (IR on, Heater On) or PoE.
- (xxiv) Horizontal Adjust Angle: 0°~360°; Vertical Adjust Angle: -90°~90°.
- (xxv) Manual Speed: 0.1°-40°/ second for Pan, 0.1°-40°/ second for Tilt.
- (xxvi) Preset speed: 40°/ second for Pan, 40°/ second for Tilt.
- (xxvii) Operating Temperature: -40°C to +60°C.
- (xxviii) Storage Temperature: -20°C to +70°C.
- (xxix) Ingress Protection: IP67.
- (xxx) Enclosure Material: 316L Stainless Steel.
- (xxxi) Humidity: 0% to 90% RH, non-condensing.
- (xxxii) Design and construction of the equipment or protective system shall have the EU-Type Examination Certification in accordance to directive 2014/34/EU or equivalent to be used in potentially explosive atmospheres and carries marking Ex db 11C T6 Gb or Ex tb IIIC T80°C Db.
- (e) Fixed Type CCTV camera minimum requirements:
 - (i) Fixed type camera for indoor/ outdoor environment.
 - (ii) Resolution: HD 1080p with 1920x1080 effective Pixels: 1920x1080 or its equivalent.
 - (iii) Video compression: H.265/H.264/MJPEG.
 - (iv) Day/Night: True Day/Night (ICR), colour/ black and white.
 - (v) Interoperability: PSIA, ONVIF, Computer Generated Imagery (CGI).
 - (vi) Edge Storage: Network Attached Storage, Local PC/ workstation for instant recoding, Micro SD Memory Slot supporting up to 128GB.
 - (vii) Web Viewer: IE10/Google/Firefox/Safari.
 - (viii) Smart Phone: IOS and Android.
 - (ix) Ethernet: RJ45 (10/100Based_T).
 - (x) Protocol: IPv4/IPv6, HTTP, HTTPS, SSL, TCP/IP, UDP, UPnP, ICMP, IGMP, SNMP, RTSP, RTP, SMTP, NTP, DHCP, DNS, PPPOE, DDNS, FTP, IP Filter, QoS, Bonjour, 802.1x
 - (xi) Power Supply: AC23V/1.5A or Power Over Ethernet (PoE) + (802.3at).
 - (xii) Power Consumption: 13W, 23W or PoE.
 - (xiii) Operating Temperature: -40°C to +60°C.
 - (xiv) Storage Temperature: -20°C to +70°C.
 - (xv) Ingress Protection: IP67.

- (xvi) Vandal Resistance: IK10.
- (xvii) Humidity: 0% to 90% RH, non-condensing.
- (xviii) Anticorrosion standard: WF2, NEMA 4X, C5-M.
- (f) Video Management and Network Recording Server minimum requirement:
 - (i) The video management and recording server shall serve as a platform, providing management of the CCTV system configuration, monitoring of overall status of all CCTV system components and reporting of equipment failures.
 - (ii) Shall be capable of continuous, scheduled, alarm/ event, and motion-triggered recording, pre and post alarm recording, and fully programmable per-channel basis.
 - (iii) The storage capacity shall be expandable, maximize storage efficiency and offer a time-and-priority-based system that identifies data to be removed when storage time expires.
 - (iv) Shall offer plug-and-play configuration and data authentication facilities.
 - (v) Shall be fully compatible with IP-based CCTV system architecture and have full over-the-network remote control and administration, system diagnostics and error logging.
- (g) System Performance
 - (i) The images shall also be clear even when camera is in full operation including when the camera is being tilted, panned, or zoomed.
 - (ii) The recorded images shall be, at all times, be accurate, sharply defined and with accurate colour reproduction under normal lighting. For reduced lighting and emergency lighting conditions, the recorded images shall minimally be accurate and defined reproduction of the scene in monochrome.
 - (iii) All monitors shall be capable to display colour images and shall possess appropriate adjustment controls (such as contrast, brightness, sharpness and colour).
 - (iv) The displayed picture on monitors shall, at all times, be sharply defined, stable, with accurate colour reproduction, and shall be free of noise, interference, ghosting and pulsing effects. Aspect ratio of the displayed picture shall be maintained with respect to the video source.
 - (v) For security system purpose, sufficient lighting shall be provided for the camera views such that the basic level of lighting shall allow camera view identification of a human face from a distance of about 10m.
- (h) Camera Tamper Protection/ Detection
 - (i) For security system purpose, the camera shall be installed in such a way that it is difficult for an intruder to change the field of view of the camera. This should be achieved by installing the camera at a suitable location/ height (minimum height of 2m from floor level) and any openings shall be secured by security fixings. Any interconnections such as cabling and antenna shall not be accessible and/or able to be torn off.

- (ii) The cameras shall be housed in vandal-resistant and tamper-proof enclosures with non-reflective, shatter-resistant glass viewing ports.
- (i) Storage Capacity
 - (i) Sufficient image recording capacity shall be provided to enable both continuous 24-hour recording of the CCTV system and archival of at least one full set of 30 days recording with an additional minimum buffer of 10%.
 - (ii) In the event of hard disk storage failure, the system shall be able to support minimum of RAID 5 array. When the bad disk is replaced by a new one, the array shall be built while the system continues to operate normally.
- (j) Basic Metadata
 - (i) The image recording equipment shall automatically record each camera ID of the camera being recorded and the date and time of the recording (synchronised to Global Positioning System (GPS) time). This information shall always be displayed where it is least likely to obscure or interfere with the image of the main subject.
- (k) Video Recording Playback
 - (i) The recording equipment shall have the features to include the capabilities to replay and normal play, still field, fast forward, rewind, record, stepping frame, visual search – forward and reverse, speed search and stop. It shall also allow fast search by date/ time slider and alarm.
 - (ii) The video footage shall be suitable for immediate playback on media player software bundled within common computer operating system (e.g. Windows Media Player and Microsoft) and/ or other open-source media player software.
 - (iii) The system shall have duplex capability or greater so as to allow simultaneous image recording, image report and playback. The system shall be designed to enable the playback of footage without causing interruption to the recording process.
 - (iv) The CCTV system shall provide for the playback of any image from any camera recorded up to 30 days previous and in a controlled environment.
 - (v) During playback, the system shall also allow variable time-control for image selection.
- (l) Image Export
 - (i) The CCTV system shall have the ability to export any image from any camera up to previous 30 days.
 - (ii) The image exported shall have no loss of individual frame quality, change of frame rate. There shall be no duplication or loss of frames in the export process. The system shall not apply any format conversion or further compression to the exported images, as this can reduce the usefulness of the content.
 - (iii) Any original metadata and/ or authentication signatures shall be exported with the images.
- (m) Replay of Exported Images

- (i) CCTV System Integrator shall provide viewer a software to allow playback or the image/ video recordings made on the CCTV system. The viewer software shall also be equipped with the capability to export recordings to open-source container and/ or common multi-media container formats such as *.avi (Microsoft), *.mov (Apple QuickTime) and *.mp4 (MPEG) to facilitate investigation work. It shall also be capable of exporting still images in *.jpeg (JPEG File Interchange Format).
- (n) Other Requirements
 - (i) All programs and software used shall be licensed. The Contractor shall provide the license, any authentication key that may come with the software and all manuals associated with the program and software at the end of the Contract.
 - (ii) CCTV system provided shall be protected against lightning induced over voltages in accordance with local codes. All metal frames of any security or CCTV consoles shall be effectively earthed and incoming power supply shall be protected by surge arrestors.
 - (iii) All equipment, except portable equipment shall be held firmly in place. This shall include cameras, mounts, cables, etc. Fastenings and supports shall be adequate to support their loads.
 - (iv) Care shall be taken in wiring to avoid damage to the cables and to the equipment. All wiring shall be executed in strict compliance with the local codes. Cables for contact switch and access card level circuits and power circuits shall run in separate conduits.
 - (v) Cables shall, as far as possible, be hooked up to equipment components via appropriate ring type cable lugs and screws or insulation displacement type terminal blocks. Where component leads/wires need to be scrapped electrically and mechanically, a wrap of turn round a terminal is considered minimum. Sufficient slack shall be allowed at connections to compensate for temperature changes and vibration. No wire shall be taut.
 - (vi) The Contractor shall test the system to show that its performance satisfies the requirement of this specification. All test equipment shall be professional and supplied by the Contractor.
 - (vii) Detailed test procedures shall be provided for commissioning. Training of operations personnel shall be included in the scope of the vendor.

G7.16.3 Access Control System

- (a) The Electronic Access Control System (ACS) shall consist of the components listed below as a minimum:
 - (i) Building access controller (BAC);
 - (ii) Electromagnetic door lock (EML);
 - (iii) Magnetic door contact (DRC);
 - (iv) Door release to exit button (DRR);
 - (v) Break glass door release (BGR); and
 - (vi) Card/ Biometric security access reader (ACR).

- (b) ACS shall be used to open/close the locks installed at specified locations and arm/disarm readers and contacts. The system shall include associated cabling, supports and other necessary ancillaries for functioning of a complete system.
- (c) Building access controller (BAC) shall be an electronic processor based controller to manage the opening and locking of the locks via card/ biometric/ keypad ACR readers. The BACS shall communicate with the ACR via Wiegand or OSDP (RS485) protocols.
- (d) The Card/ Biometric security access reader shall have the following features and specifications:
 - (i) Rugged structure for outdoor installation and extra durability - IP65 rated waterproof and weatherproof structure.
 - (ii) The Card/ Biometric security access reader (ACR) shall be of contactless smart card or fingerprint reader or a keypad reader.
 - (iii) For card reader, the ACR reader shall be of SS 529 SSID PS21 Card platform and shall be able to support Wiegand, RS232 and OSDP (RS485) protocols.
 - (iv) For biometric ACR, reader shall provide a high definition of optical fingerprint sensor with a verification speed of less than 1 second.
 - (v) All ACR readers shall be integrated with the site wide ISMS.
 - (vi) For biometric ACR, the reader shall support 1:1 authentication mode based on Card+Fingerprint authentication and 1:N authentication based on Fingerprint Only.
 - (vii) The reader module shall have a minimum processing speed of 400MHz DSP and minimum memory of 4MB flash + 8MB RAM.
 - (viii) The biometric ACR reader shall be able to meet the following performance criteria:
 - 1) False Accept Rate of not more than 0.00001%.
 - 2) False Reject Rate + Failure to Acquire Rate or not more than 0.01%.
 - (ix) Technical Specifications:
 - 1) The ACR reader shall have the following modes of operation as a minimum:
 - Card Only;
 - Fingerprint Only;
 - Card + PIN Only;
 - Fingerprint + PIN only; and
 - Card + Fingerprint + PIN.
 - 2) For ACR keypad reader, the keypad should have a 3 X 4 matrix keypad. The keypad should have a backlight to allow for easy viewing.
 - 3) The ACR keypad reader with display shall have a 16 characters X 2 lines alphanumeric Organic LED Display.

- 4) The ACR readers' enclosure shall be secured to prevent any unauthorised opening and complete with tamper switch to alert operator.
- 5) For biometric ACR, the module shall be able to meet the following certifications and standards as a minimum:
 - ISO/IEC 19794-2 data standard;
 - FBI PIV /FIPS 201 Certified;
 - NIST certified interoperable template/image format standards (ISO 19794-2/ISO 19794-4 /ANSI -378);
 - Image Compression Standard (WSQ); and
 - CE, FCC, KCC, WHQL, CB, USB-IF, RoHS, WEEE.
- (e) For card ACR, the reader shall read the access ID card belonging to authorized Board's employees. The reader shall send record to the system of the entry / exit information including date and time. Alarm shall be activated for any unauthorized access.
 - (i) The card ACR reader shall have a local indicator which is normally red, momentarily changes to green when it has been confirmed that ID card read by the reader is authorised for use at that location and entry will subsequently be given to the said location.
 - (ii) The response time to unlock a door shall not be more than one (1) second, with a tolerance of +/- of 0.5 sec.
 - (iii) The overall thickness of the door access ACR reader unit shall not exceed 30 mm.
 - (iv) No external wiring to the ACR reader shall be allowed to activate the door lock. Removal of the ACR reader from its mounted position activates a tamper alarm.
 - (v) Doors which require access via card/ biometric readers will be furnished with an EML (electromagnetic lock) together with a DRC (magnetic door contact) to indicate door position. Doors without such access requirements or key lock will be furnished with a magnetic contact only.
 - (vi) Each ACR reader shall be able to hold at least unique identities corresponding to staff and authorised visitors to the Plant. Identity quantities will be as per the Particular Specifications.
 - (vii) Any modification function to the employee data shall be available at the system and function is user-friendly.
- (f) Door Release Points (DRR)
 - (i) Door release to exit button (DRR) shall be surface mounted for access control application.
 - (ii) Door release to exit button (DRR) shall be waterproof and suitable for outdoor application if provided for outdoor locations; else, it shall be suitable for the installation locations as specified. It shall be quick and easy to install with 'plug and play' concept to minimize wiring time. Door release point shall be mounted on secured side or the inside of the room.

- (iii) A BGR (break glass door release) shall be installed beside each release to exit button for emergency exit.
- (g) Break Glass Door Release (BGR)
 - (i) Provide monitored emergency Break Glass Door Release (BGRs) where necessary for emergency escape purposes on the secure side of controlled doors.
 - (ii) Install flush-mounted BGRs wherever possible. BGRs shall be engraved "EMERGENCY DOOR RELEASE" or similar wording to be agreed with the S.O. Colours shall be approved by the S.O..
 - (iii) Operation of the unit shall release the associated door lock such that the door may be freely used. The BGR shall operate directly on the lock power supply line to ensure fail-safe exit capability.
 - (iv) Activation of any emergency break glass door release or attempt at removal of the glass shall activate an alarm on the workstation alarm display.
 - (v) Provide a total of 20% spare glass panels.
- (h) Electromagnetic Locks (EML)
 - (i) The EML lock shall be made of stainless steel or corrosion-resistant metals, tamper resistant and heavy-duty construction. It shall be of fail safe operation (power to lock). The door contact shall be designed to suit most doors, with attractive finished and coated with chrome or anodised aluminium equivalent, rust resistant attractive metal. The EML lock shall have a high holding forces suitable where it is installed for the door with instant fail unlock for maximum safety and optional door monitoring option. The EML lock shall operate silently.
 - (ii) All door contacts shall be mounted onto doors with rigid mounting template and fixings. Screws should be countersunk. Door contact shall be rated to operate in Singapore weather conditions, with low residual magnetism immediately after switch off. It should be powered by voltage not higher than extra low voltage.
 - (iii) Cabling into and out of door contacts should be well concealed or armoured if installed outdoor.
 - (iv) Every door fitted with EML lock shall be provided with automatic door closers. Each automatic door closer shall come with quick fix template for fast and accurate fitting onto the door.
 - (v) Each door fitted with EML lock shall be provided with a magnetic door contact (DRC) which indicates the open and close status to the plant security system.
 - (vi) The Contractor shall provide force dispersing corrosion resistant metal plate onto door or door frames if they are found to be hollow or weak due to wear and tear, weathering or damage.
- (i) Fire Interface Requirements
 - (i) Provide an interface from the fire alarm system to ensure free egress under emergency situations of all access controlled doors unless specifically indicated otherwise.

- (ii) Install connections to any fire alarm interfaces where the form of the lock used demands it.
- (iii) Ensure that associated doors release on the receipt of the fire alarm by direct interruption of the lock power. Include for any additional switching required, together with the relevant lock power distribution.
- (iv) Provide interfaces for all door controllers forming part of the access control system and additionally for any other doors which have electric locking via local or remote control.

G7.16.4 Fence Intrusion Detection System (FIDS)

- (a) The system shall include, but not limited to intrusion sensors, gate contacts, active beam sensors, sensor cabling, signal processing units, data/power cabling including interfaces to a Guard House and Control Room or otherwise specified location for remote monitoring and for activation of the Site Security and CCTV Surveillance System. All of the perimeter fence shall be covered with fence intrusion detection sensors including all gates and fence openings.
- (b) When an alarm is triggered by the FIDS, it shall activate the CCTV camera covering the respective zone to pop up to the designated CCTV monitor and to sound a siren and strobe light (adjustable volume). The design the zoning of the fencing shall be done together with the Site Security CCTV Surveillance System.
- (c) FIDS shall be an electronic perimeter intrusion detector comprising fibre optic cable system directly attached to the fence using ultraviolet resistant cable tie at specified/ indicated locations and connected to single and/or dual zone electronic signal analysers.
- (d) The FIDS shall be controlled by an independent microcontroller with control and monitoring interfaced with a Monitoring Workstation(s) via TCP/ IP Ethernet.
- (e) For hazardous locations, all controllers, equipment, installation and fitting accessories, etc shall be intrinsically safe or explosion proof.
- (f) The FIDS shall also be protected from lightning and shall be capable of operating in extreme EMI (electromagnetic interference) and RFI (radio frequency interference) environments at site with no effect on the normal operating characteristics.
- (g) The system shall be provided with surge protection devices to protect each control unit and equipment against transient voltage surge caused by lightning and from the power supplies system.
- (h) The FIDS shall be capable to detect any unauthorised entry by intruders with intent to penetrate through the perimeter fences and gates by climbing or cutting through or any kind of intrusion through the perimeter fencing and fence gates. The FIDS shall be provided with anti-tampering features. An alarm shall be activated in an event of tampering or when the FIDS, is switched off without proper disarming procedures.
- (i) The monitoring location shall be able to perform the following remote-control operations from the workstation via TCP/IP Ethernet but not limited to:
 - (i) Arm/Disarm the FIDS zone(s);
 - (ii) Silent the FIDS audible alarm;

- (iii) Acknowledge the FIDS zone alarm status; and
- (iv) Monitor the FIDS zone, battery and AC status.
- (j) After FIDS is activated, the FIDS zone shall be displayed to indicate the location of alarm in the Monitoring workstation. If alarm is activated in the monitoring workstation, an audible alarm shall be sounded. The monitoring system software shall highlight the exact plan/zone of the alarm activation to alert the security operators.
- (k) The FIDS shall be able to operate independently in case of any failure from the Monitoring Workstation(s) or LAN/ WAN communication interface. The output status shall be displayed on the Monitoring workstation(s) in real-time, on the graphics display. The command shall not interfere with the normal operation of various sub system monitors and controllers.
- (l) When there is an intrusion/ violation/ tampering and false alarm(s) is activated, the operator shall assess the nature of the intrusion/ violation/ tampering, and carry out reset/ bypass/ arm, disarm/ changing of zone characteristics parameter or even change the user password via Monitoring Workstation(s).
- (m) Where FIDS sensor cables or communication network cables leave the fence and enter into the ground, they must be protected by stainless steel SS304 conduits to prevent accidental cutting. Equipment used outdoors shall be weatherproof and waterproof with min of IP65 rating.
- (n) All cablings shall run along fence perimeter, and through the road where necessary using road crossings cable ducts. Road crossings shall be protected by HD uPVC conduit pipe. All cables running along the perimeter fence shall be neatly secured together into the stainless steel or corrosion resistant cable trunkings mounted on the inside along the concrete base of the fence. No cables shall run within building/or underground of areas not belong to the Board.
- (o) If specified in the scope of works to refurbish existing Site Security CCTV camera junction boxes, tamper-proof built-in locking mechanism shall be provided and the junction boxes shall be encased in die-cast aluminium or stainless-steel casing to minimum of IP55 rating.

G7.16.5 Centralised ISMS Server and Monitoring Equipment

- (a) A centralised database server and system software shall be provided to integrate and manage multiple BACSS, ACRs, security access locks, site security CCTV system and FIDS.
- (b) ISMS software and centralised database shall be based on open platform communication.
- (c) Monitoring workstation(s) shall be provided to access the control and monitoring of ISMS.
- (d) Hardwire interlocks connection shall be provided through central database ISMS between ISMS locks and fire alarm system where the power supply shall cut off and door locks shall open automatically if fire alarm is triggered in the associated security access zone.

END OF SECTION